PROJECT REPORT

Vendor Performance Analysis

The project mainly focuses on analyzing vendor performance, sales trends, and profitability. It aims to identify key vendors and brands contributing to revenue while highlighting underperforming areas that require improvement. The analysis also emphasizes optimizing inventory management by reducing unsold stock and improving turnover efficiency. Additionally, it explores cost optimization strategies such as bulk purchasing to enhance overall business profitability and decision-making.

Tools Used-

- **Python (Pandas) for data analysis**
- **Matplotlib & Seaborn for visualization**
- **Power BI for dashboard creation**

BUSINESS PROBLEM

Effectively managing vendor performance, inventory, and profitability. Despite generating significant sales, a portion of capital remains tied up in unsold inventory, indicating inefficiencies.

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Data Ingestion

Handling a very large dataset consisting of crores of records, which could not be loaded directly into memory. To efficiently manage this, a SQLite database was created and all raw data files were ingested into it. Using SQL queries and multiple tables were combined, filtered, and only the necessary columns were extracted. Finally, a single consolidated summary table was created, which served as the base dataset for further analysis.

- Handled large dataset using SQLite
- Ingested raw files into database
- Used SQL to merge tables
- Created a single summary table

Data Cleaning

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).
- Removed null values using dropna() to avoid incorrect calculations.
- Replaced infinite values (inf, -inf) with nulls and removed them.

Issues Faced

During the analysis, incorrect unsold inventory values were observed due to negative inventory, where sales exceeded purchases. This occurred because older stock was being sold, and the issue was resolved by converting negative unsold units to zero for accurate calculations.

- Negative inventory issue
- Sales exceeded purchases due to old stock
- Converted negative unsold units to zero

Metrics Created

To transform raw data into meaningful insights that support better decision-making. They helped in evaluating vendor performance, profitability, and inventory efficiency in a more structured and measurable way.

- Gross Profit, Profit Margin
- Stock Turnover
- Sales to Purchase Ratio
- Unsold Inventory

Key Findings

- A significant portion of total sales (around 65%) is driven by a small number of top vendors, indicating a dependency risk.
- Several brands have high profit margins but low sales volume, showing strong potential for targeted promotion.
- Bulk purchasing was found to reduce the per-unit cost, helping improve overall profit margins.
- Some inventory remains unsold, tying up capital and highlighting the need for better inventory planning.
- There is a clear performance gap between top-performing and low-performing vendors in terms of sales and profitability.

Dashboard

- The dashboard provides a comprehensive overview of key KPIs such as total sales, profit, purchase, unsold capital, and profit margin (%).
- It includes bar charts to visualize top-performing vendors and brands based on sales and profitability.
- A donut chart highlights the contribution of top vendors to overall purchases, helping identify dependency.
- Scatter plots are used to identify target brands based on sales and profit margin for strategic decision-making.
- A funnel chart is used to highlight low-performing vendors, helping to easily identify areas that require improvement and strategic focus.

Business Insights

- A major portion of sales (~65%) is generated by a few top vendors, indicating high dependency risk.
- Several brands have high profit margins but low sales, representing strong growth opportunities through promotion.
- Bulk purchasing significantly reduces per-unit cost, improving overall profitability.
- Presence of unsold inventory shows inefficient inventory planning and capital blockage.
- There is a noticeable gap between high-performing and low-performing vendors in terms of sales and profit contribution.

Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

Final Point

This project helped me understand how to handle large-scale data efficiently, from data ingestion using databases to cleaning and transforming it for analysis. It enhanced my ability to derive meaningful business insights by evaluating vendor performance, profitability, and inventory efficiency. Additionally, I gained practical experience in data visualization and dashboard creation, enabling better decision-making and strategic planning in real-world business scenarios.